When More Modalities Hurt: Conflict-Aware Evaluation and Fusion for Multimodal Sentiment Models

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Abstract

Adding modalities is widely assumed to improve predictive models. We study multimodal sentiment classification on Amazon Reviews 2023, combining review text (SBERT embeddings) with structured product/review metadata, and ask a narrower question: *when modalities disagree, does fusion still help?* Using a fixed, balanced, leakage-controlled evaluation protocol with redrawn splits across five seeds, we observe that early fusion improves average accuracy by only a small but significant margin over a strong text-only baseline (+0.5 points overall, paired t-test p=0.002), while *worsening* the calibration gap between agreement and disagreement cases (+0.069, p=0.002). On the disagreement subset itself, fusion provides no significant accuracy gain (p=0.79). We show the phenomenon persists across five product categories and is not an artifact of product-level rating priors: removing leakage-prone features (`product_average_rating`, `product_rating_number`) significantly lowers accuracy (−0.8 points, p=0.024), confirming those features inflate results rather than explaining the conflict failure. We introduce **UGCA-Fusion** (Uncertainty-Guided Conflict-Aware Fusion), a learned method whose conflict detector estimates modality conflict at inference *without the true label* (AUROC ≈0.72). UGCA-Fusion's conflict detector is genuinely predictive and its components ablate sensibly, but in our setting it does not significantly beat early fusion or simple disagreement-aware reweighting on disagreement accuracy. Our main claim is deliberately cautious: *average multimodal gains mask conflict-specific reliability failures, and conflict-aware inference-time fusion can reduce, but not eliminate, these failures.*

1 Introduction

Multimodal models are typically evaluated by average metrics, which can hide systematic failures on instances where modalities provide conflicting evidence. In product reviews, a glowing text paired with a low star rating (or vice versa) is a concrete instance of *modality conflict.* We ask whether the standard practice of fusing text and metadata helps or hurts precisely on these conflict cases, and whether any apparent benefit survives (i) leakage control, (ii) multiple seeds, and (iii) multiple product domains.

Our contributions, stated cautiously and supported by multi-seed evidence:

1. A systematic, leakage-controlled evaluation protocol for modality conflict with a full conflict-aware metric suite (Section 3).

2. Evidence across five Amazon categories that fusion's average gains coincide with a *widening* agreement/disagreement calibration gap (Sections 7, 8).

3. A leakage-controlled analysis showing the phenomenon is not caused by product-level rating priors (Section 7).

4. **UGCA-Fusion**, a learned inference-time conflict-aware fusion method with a label-free conflict detector (Section 5).

5. Calibration and statistical analyses, plus a human-annotation pipeline that separates genuine conflict from label noise (Sections 9, 10).

We are explicit about negative results: UGCA-Fusion does not solve conflict, and we report it as such.

2 Related Work

Multimodal fusion has a long history [Baltrušaitis et al., 2018, Zadeh et al., 2017], with evidence that naively adding modalities can hurt optimization and generalization [Wang et al., 2020, Peng et al., 2022]. Trusted/uncertainty-aware multi-view methods weight modalities by estimated reliability [Han et al., 2021, Kendall et al., 2018]; our conflict detector is related but targets *instance-level* conflict inferred from unimodal predictive distributions. Calibration of modern networks is well-studied [Guo et al., 2017, Naeini et al., 2015], as is selective prediction [Geifman and El-Yaniv, 2017]; we apply these tools *conditioned on conflict*. For evaluation we use McNemar's test [Dietterich, 1998] and bootstrap confidence intervals. Cross-modal sentiment benchmarks such as CMU-MOSEI [Zadeh et al., 2018] and Hateful Memes [Kiela et al., 2020] are natural future testbeds (Section 12).

3 Problem Setup

We use Amazon Reviews 2023 [Hou et al., 2024]. For each category we build a fixed balanced pool of 20,000 reviews (positive: stars ≥ 4; negative: stars ≤ 2; stars = 3 dropped), encode text with SBERT (`all-MiniLM-L6-v2`) [Reimers and Gurevych, 2019], and form metadata features. The pool is fixed with a master seed; only the stratified 70/15/15 train/val/test split is redrawn per run seed ($\{42, 123, 456, 789, 2026\}$), exposing split-level variance while keeping the expensive encoding to a single CPU pass.

Leakage control. The full metadata set includes product-level aggregates (`product_average_rating`, `product_rating_number`) that can encode the review label. Our *main* setting is **leakage-controlled (LC)**: these aggregates are removed. We report full-feature results only as a contrast.

Metric suite. Beyond accuracy/F1/AUROC we report ECE, Brier score, accuracy and ECE *stratified* by agreement/disagreement/strong-disagreement, the *calibration gap* (disagreement ECE − agreement ECE), high-confidence error rate, confidence on correct vs. incorrect predictions, and modality dominance. We use bootstrap 95% CIs, paired bootstrap tests, and McNemar's test.

4 Disagreement and Conflict Definitions

Because "disagreement" is not uniquely defined, we compare four definitions on the All_Beauty pool using 5-fold cross-fitted (leakage-free) predictions: **(A)** pretrained-sentiment (DistilBERT-SST-2 [Sanh et al., 2019]) vs. rating label; **(B)** cross-model disagreement between text-only and metadata-only (LC) predictions; **(C)** high-confidence text contradiction (text confidence ≥ 0.90 and text prediction $\neq$ label); and **(D)** a 200-example human-annotation subset.

Table 1: Disagreement definitions on All_Beauty (cross-fitted). Low Jaccard overlap indicates the definitions capture different instances.

Definition	Prevalence	MM acc (flagged)	MM acc (unflagged)
A: sentiment vs. label	11.6%	0.638	0.912
B: cross-model	42.3%	0.859	0.896
C: high-conf contradiction	1.3%	0.008	0.892
Jaccard: A∩B= 0.11, A∩C= 0.07, B∩C= 0.01			

Table 1 shows the definitions flag *largely disjoint* instances (pairwise Jaccard ≤ 0.11), so the choice of definition materially changes conclusions. Definition A reproduces a large multimodal failure (accuracy 0.64 on flagged vs. 0.91 elsewhere); Definition C is rare but the model is almost always wrong on it (0.008), consistent with label noise / sarcasm—motivating human annotation (Section 10). Unless noted, the rest of the paper uses Definition A.

5 UGCA-Fusion

Uncertainty-Guided Conflict-Aware Fusion aims to know, at inference and without the true label, when modalities conflict, and to route fusion accordingly. Text and metadata encoders produce $h_{\text{text}}, h_{\text{meta}}$ and unimodal heads produce $p_{\text{text}}, p_{\text{meta}}$. A conflict detector predicts $\hat{c}$ from conflict features: $|p_{\text{text}} - p_{\text{meta}}|$, the entropies of p_{text} and p_{meta}, and the Jensen–Shannon divergence [Lin, 1991] between the unimodal distributions (plus optional learned features). A gate g, modulated by $\hat{c}$, forms $h_{\text{fused}} = g\, h_{\text{text}} + (1-g)\, h_{\text{meta}}$, and a classifier predicts $\hat{y}$. The training loss is

$$\begin{aligned}\mathcal{L} = \mathrm{CE}(\hat{y}, y) + \lambda_{\text{t}}\mathrm{CE}(p_{\text{text}}, y) + \lambda_{\text{m}}\mathrm{CE}(p_{\text{meta}}, y)\\ + \lambda_{\text{c}}\,\mathrm{BCE}(\hat{c}, d) + \lambda_{\text{cal}}\,\Omega_{\text{cal}},\end{aligned}$$

where d is the proxy disagreement label (used *only in training*) and Ω_{cal} is a differentiable calibration surrogate. At inference the detector uses no label, which is the core claim we test empirically.

6 Experimental Setup

Text-only is SBERT+logistic regression; metadata-only is XGBoost; fusion models are MLPs (early/late/gated). Disagreement-aware reweighting upweights training disagreement rows (w=5). UGCA-Fusion and four ablations (no conflict detector, no entropy, no JS, no calibration loss) train for up to 25 epochs with early stopping. All models share each seed's split. Compute is CPU-only; this constrains the second-dataset scope (Section 12).

7 Main Results

Table 2 reports the canonical All_Beauty results (mean±std over five seeds; pool disagreement rate 11.6%). Three observations hold:

1. **Fusion's overall gain is small.** Early-fusion (LC) accuracy 0.881 vs. text-only 0.876 (+0.005; paired t-test p=0.002, Cohen's d_z=3.2). On the disagreement subset the gain is not significant (+0.002, p=0.79).

Table 2: Canonical All_Beauty, mean±std over 5 seeds. LC = leakage-controlled (main). Calibration gap = disagreement ECE − agreement ECE (higher is worse).

Model	Accuracy	Disagree. acc	Strong-dis. acc	AUROC	ECE	Calib. gap
Text-only	0.876±0.005	0.652±0.025	0.619±0.031	0.944	0.031	0.120
Metadata-only (LC)	0.578±0.007	0.508±0.022	0.485±0.017	0.610	0.040	0.113
Metadata-only (full)	0.686±0.004	0.628±0.014	0.619±0.018	0.758	0.027	0.099
Early fusion (LC)	0.881±0.005	0.654±0.032	0.626±0.038	0.947	0.038	0.189
Early fusion (full)	0.889±0.007	0.678±0.040	0.658±0.043	0.955	0.041	0.188
Late fusion (LC)	0.884±0.005	0.650±0.038	0.622±0.054	0.952	0.021	0.181
Gated fusion (LC)	0.883±0.004	0.651±0.024	0.625±0.032	0.951	0.018	0.168
Disagree-aware (LC)	0.867±0.004	0.678±0.025	0.660±0.032	0.937	0.060	0.170
UGCA-Fusion (LC)	0.883±0.006	0.658±0.035	–	0.95	0.050	0.206

Table 3: Cross-domain (mean over 3 seeds). EF = early fusion (LC), TO = text-only.

Category	Acc (TO/EF)	Disagree (TO/EF)	Calib. gap (TO/EF)
All_Beauty	0.873/0.878	0.636/0.636	0.132/0.207
Appliances	0.861/0.871	0.644/0.662	0.177/0.231
Digital_Music	0.876/0.886	0.596/0.636	0.108/0.190
Gift_Cards	0.914/0.924	0.568/0.646	0.249/0.263
Video_Games	0.870/0.872	0.603/0.586	0.158/0.253

2. **Fusion worsens conflict calibration.** The calibration gap rises from 0.120 (text-only) to 0.189 (early-fusion LC); the per-seed paired increase is +0.069 (p=0.002, d_z=3.2). Gated/late fusion improve *average* ECE but the conflict gap remains elevated.

3. **Not explained by leakage.** Full-feature early fusion beats LC by +0.8 points accuracy (p=0.024) and +2.4 points disagreement accuracy (p=0.011): the product-rating aggregates inflate results, so the LC setting (our main one) is the honest baseline.

8 Cross-Domain Generalization

Across five categories (All_Beauty, Appliances, Digital_Music, Gift_Cards, Video_Games; seeds $\{42, 123, 456\}$), early fusion (LC) consistently increases *both* average accuracy *and* the calibration gap relative to text-only (Table 3). Pooling over category×seed, the accuracy gain is +0.7 points (p=0.0006) and the disagreement-accuracy gain is modest (+2.4 points, p=0.043); notably the disagreement effect is inconsistent in sign (e.g. Video_Games: 0.586 vs. 0.603). The calibration-gap increase, by contrast, is consistent across all five categories. We therefore claim replication for the *calibration* failure but only weak, domain-dependent support for a disagreement-accuracy benefit.

Scope of cross-category results. The cross-category figures and Table 3 include text-only, metadata-LC, early-fusion-LC, and disagreement-aware reweighting. UGCA-Fusion (Section 5) is evaluated separately on the canonical All_Beauty 5-seed benchmark (Table 2) and is *not* included in the cross-category figures, to avoid mixing complete and incomplete category coverage. Extending UGCA-Fusion to all categories and seeds is left to future work (Section 12).

9 Calibration, Selective Prediction, and UGCA Analysis

The conflict-aware metric suite shows fusion concentrates its calibration error on conflict cases. UGCA-Fusion's *conflict detector* predicts the proxy disagreement label on the test set with AUROC $\approx$0.72 (mean over 5 seeds) *using no true label*, supporting the core hypothesis that conflict is detectable at inference. Ablations behave as designed: removing the conflict detector lowers disagreement accuracy (0.658$\rightarrow$0.646) and removing the calibration loss worsens the calibration gap (0.206$\rightarrow$0.221). However, across five seeds UGCA-Fusion is statistically indistinguishable from early fusion on accuracy and on the disagreement subset (paired p>0.7; example-level McNemar median p>0.5), and gated fusion remains best-calibrated (ECE 0.018). UGCA-Fusion does significantly beat disagreement-aware reweighting on *overall* accuracy (+1.6 points, p=0.001), as reweighting trades overall accuracy for disagreement accuracy.

10 Human Annotation / Qualitative Analysis

We provide a reproducible human-annotation pipeline: a six-category guide (genuine conflict, label noise, mixed sentiment, sarcasm, unclear, metadata-prior), a batch generator, and an analyzer computing the genuine-conflict vs. label-noise breakdown, Cohen's κ inter-annotator agreement, and model accuracy per human category. We export 200 cross-fitted disagreement candidates for labeling. We *do not* fabricate labels; the analyzer reports "annotation pending" until human files exist. This separates genuine model failures from dataset artifacts and is a stated prerequisite for strong claims about "conflict."

11 Ablations

See Section 9 and `reports_icml/tables/ugca_ablation.csv`. Each UGCA component (conflict detector, entropy features, JS divergence, calibration loss) contributes in the expected direction, but the marginal effects are small relative to seed variance.

12 Limitations

(1) The disagreement label is a *proxy*; Section 4 shows definitions disagree, and human annotation is pending. (2) Both evaluated families are review+metadata; a second *family* is scoped (Yelp; feasibility report included) and a true cross-modality test (CMU-MOSEI, Hateful Memes) is GPU-required future work. (3) Compute is CPU-only, so encoders are frozen and pools are capped at 20k. (4) With five seeds, p-values are interpreted cautiously alongside effect sizes and CIs. (5) UGCA-Fusion is a diagnostic advance, not an accuracy solution.

13 Conclusion

Across the evaluated categories we observe that average multimodal gains mask conflict-specific reliability failures: fusion barely improves overall accuracy, does not reliably help on disagreement cases, and significantly worsens the agreement/disagreement calibration gap, even after leakage control. Our UGCA-Fusion can *detect* conflict at inference without labels but does not eliminate the failure. We therefore frame robust multimodal behavior under modality conflict as an open problem, and release a leakage-controlled, multi-seed, multi-category evaluation protocol to study it.

Reproducibility. All tables and figures are regenerated by the `scripts_icml/` pipeline; per-experiment provenance is logged in `reports_icml/experiment_registry.md`.

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